Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Brief Study Title: Study of the Efficacy and Safety of Various Anti-inflammatory Agents in Participants With Mild Cognitive Impairment or Mild Alzheimer's Disease **Full Study Title:** Exploratory PLatform Trial on Anti-Inflammatory Agents in Alzheimer's Disease (EXPLAIN-AD): A Randomized, Placebo-controlled, Multicenter Platform Study to Evaluate the Efficacy, Safety, Tolerability and Pharmacokinetics of Various Anti-inflammatory Agents in Patients With Mild Cognitive Impairment Due to Alzheimer's Disease and Mild Alzheimer's Disease **Short Title/Acronym:** EXPLAIN-AD **Protocol Number:** CADPT06A12201 **NCT Number:** NCT04795466 **Posting ID:** 342862574179924281 **Sponsor Name:** Novartis (Novartis Pharmaceuticals) **Trial Status:** TERMINATED (Study was terminated early) **Investigational Product:** Canakinumab (ACZ885) | **Trade Name:** Ilaris | **ATC Code:** L04AC08 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/ilaris-epar-product-information_en.pdf **Phase of Development:** Phase 2 (Phase IIa) **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of anti-inflammatory agents (Canakinumab) on cognition in early Alzheimer's disease (mild cognitive impairment or mild AD). **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics, as well as their effects on central and peripheral inflammation.

2.2 Background & Rationale To address the role of neuroinflammation in the progression of early Alzheimer's disease. There is increasing evidence that inflammation in the brain and nervous system is a key contributor to the development of AD. This study aims to determine whether anti-inflammatory agents like Canakinumab (an IL-1β blocker) can have a safe and beneficial effect on memory and thinking abilities, as well as on established biomarkers of the disease.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant- and Investigator-blinded) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target Enrollment (N): 34 participants (Trial terminated early; initially planned as a continuous platform study, but only one experimental arm was enrolled.) **Number of Sites:** 10 locations (including sites in the US, UK, and Finland) **Estimated Study Start:** 28-Oct-2021 **Estimated Primary Completion:** 07-Mar-2024

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female, age ≥ 45 years and ≤ 90 years at the time of signing the informed consent. 2. Participant has a reliable study partner or caregiver who can accompany the participant to all visits. 3. A diagnosis of probable MCI due to AD or mild AD according to the National Institute on Aging and the Alzheimer's Association (NIA-AA) criteria. 4. Confirmed amyloid and tau positivity via CSF sampling performed at screening. 5. Mini-Mental State Examination (MMSE) total score of 20 to 24 (inclusive) at screening; OR, MMSE total score of 25-30 (inclusive) plus a DSST score at least 0.5 standard deviation (SD) below normative data at screening.

4.2 Exclusion Criteria 1. Use of an investigational agent or an approved product with the intent to modulate inflammation or modulate the course of AD (e.g., Tau ASOs, gene therapy, amyloid or tau vaccine). * Previous use of small molecules is allowed if discontinued for at least five half-lives, or at least 30 days from when the expected pharmacodynamic effect has returned to baseline prior to screening, whichever is longer. * Previous use of monoclonal or polyclonal antibodies or other biologics is allowed if discontinued for at least five half-lives prior to screening. 2. Current medical or neurological condition that might impact cognition or performance on cognitive assessments, e.g., MCI not due to AD, non-Alzheimer dementia, Huntington's disease, Parkinson's disease, stroke, schizophrenia, bipolar disorder, active major depression, multiple sclerosis (MS), amyotrophic lateral sclerosis (ALS), active seizure disorder, or history of traumatic brain injury associated with loss of consciousness and ongoing residual transient or permanent neurological signs/symptoms including cognitive deficits, and/or associated with skull fracture. 3. Diagnosis of vascular dementia prior to screening (e.g., modified Hachinski Ischaemic Scale score > 6 or those who meet the NINDS AIREN criteria for vascular dementia).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Canakinumab 150 mg once every 4 weeks for the first 2 doses, followed by 300 mg once every 4 weeks for the subsequent 4 doses (or matching placebo). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 20 weeks (Day 1 to Day 141).

5.2 Concomitant & Prohibited Therapy Permitted: Previous use of small molecules is allowed if discontinued for at least five half-lives or 30 days prior to screening. Participants could continue taking certain approved baseline medicines for Alzheimer's disease. **Prohibited:** Investigational agents or approved products intended to modulate inflammation or the course of AD.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -60 to -8	Informed Consent, CSF sampling, MMSE, Biomarker screening
Baseline	Day -7 to -1	Randomization, Baseline cognitive assessments
Treatment Period	Day 1 to 141	SC Injection (Q4W), Safety Labs, NTB Assessments
End of Cohort 1 (EOC1)	Day 171	Primary Endpoint Evaluation, NTB Z-Scores (approx. 30 days after last dose)
End of Cohort 2 (EOC2)	Day 281	Final Follow-up, Safety Evaluation (approx. 140 days after last dose)

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Cognition as Measured by the Neuropsychological Test Battery (NTB) Z-scores at Day 171. **Measurement Method:** NTB composite consisting of RAVLT (immediate/delayed), Wechsler Memory Scale Digit Span, COWAT, and CFT. Raw scores are converted to z-scores and averaged. A zero Z-score means no cognitive change, a negative value indicates decline, and a positive value means improvement.

7.2 Secondary & Exploratory Endpoints * Change From Baseline in Memory as Measured by the Total Composite NTB Memory Z-score at Day 171. * Change From Baseline in Executive Function as Measured by the Total Composite NTB Executive Function Z-score at Day 171. * Change From Baseline in Digit Symbol Substitution Test (DSST) Score - CANTAB. * Relative % change from baseline in volume of distribution (Vt) of the radio tracer for TSPO after treatment (PET TSPO - marker of central inflammation). * Change in Neuropsychiatric Inventory (NPI) total score.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the 20-week treatment period and extending to the 281-day safety follow-up. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for clinical trial safety data. **Data Monitoring Committee (DMC):** Utilized for ongoing safety evaluation during the platform study progression.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE monitoring. **Sample Size Justification:** As an exploratory platform trial, power was estimated for proof-of-concept cognitive shifts, though the study was terminated early resulting in only one experimental arm being evaluated (Targeted Enrollment = 34). **Handling of Missing Data:** Standard multiple imputation/mixed-effects models for repeated measures to handle missing NTB components.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite clinical monitoring combined with eNeuropsychiatric at-home assessments. **Audit Trail:** eCRF entry tracking, secure biomarker/CSF sample chain of custody, and verified clinical dementia rating procedures.

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11. Study Identifiers & Governance

Primary ID: CADPT06A12201 **Registry Number:** NCT04795466 **Posting ID:** 342862574179924281 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Brief Title:** Study of the Efficacy and Safety of Various Anti-inflammatory Agents in Participants With Mild Cognitive Impairment or Mild Alzheimer's Disease **Official Regulatory Title:** Exploratory PLatform Trial on Anti-Inflammatory Agents in Alzheimer's Disease (EXPLAIN-AD): A Randomized, Placebo-controlled, Multicenter Platform Study to Evaluate the Efficacy, Safety, Tolerability and Pharmacokinetics of Various Anti-inflammatory Agents in Patients With Mild Cognitive Impairment Due to Alzheimer's Disease and Mild Alzheimer's Disease

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12. Clinical Framework (Alzheimer's Disease Specific)

Primary Conditions/Disease Areas: Mild Cognitive Impairment, Alzheimer Disease **Diagnosis Threshold:** Confirmed amyloid and tau positivity via CSF sampling; MMSE 20-30 with functional decline (or DSST considerations) **Medical Methodology:** Anti-inflammatory Disease Modifying Therapy **Optimization Target:** Central and peripheral inflammation reduction (IL-1β inhibition)

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13. Study Procedures & Methodology

Management Pathway: Screening via CSF/PET and cognitive testing **Education Protocol (HCP):** Site-specific training for proper NTB and MMSE administration **Education Protocol (Patient):** Caregiver/ study partner requirement and eNeuropsychiatric at-home assessment instructions **Quality Audit Mechanism:** Centralized rater scoring review for primary cognitive endpoints

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14. Observation & Follow-Up Schedule

Total Duration: Up to 281 days (~40 weeks) **Onsite Visit Cadence:** Every 4 weeks during the 20-week treatment period, plus EOC1 (Day 171) and EOC2 (Day 281) **Remote Monitoring Frequency:** eNeuropsychiatric at-home assessments conducted frequently between clinic visits **Checklist Review Interval:** Routine safety checks every 4 weeks

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]